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SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Duration: 60 Minutes

Total Marks: 20

Code of Conduct

Shanika (192511387)

I Betty A. (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Betty

Criteria	Concept Identification (1.25)	Relationship Mapping (1.25)	Hierarchy and Organization (1)	Technical Relevance (1)	Visual Clarity (0.5)	TOTAL (5)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Architecture: Include front-end, back-end, client devices, data centers, and distributed servers.
2. Concept Map 2: Characteristics of Cloud Computing: Connect on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.
3. Concept Map 3: Deployment Models: Map public cloud, private cloud, hybrid cloud, and community cloud with usage scenarios.
4. Concept Map 4: Cloud Benefits vs Challenges: Show advantages (cost efficiency, scalability, flexibility) and limitations (security, downtime, vendor lock-in).

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

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Cloud Computing Architecture

consist of

Front-end
(clients)

consists of

Back-end
cloud infrastructure

assess via

Internet /
network

send
request

manages

Data Centers

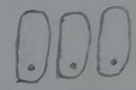
contain

Distributed
servers

Storage
system

Laptop
Smartphones
Tablets
Desktops

Servers
virtualization
Storage
networking
management
services



Characteristics of cloud computing

On-Demand self-service

means

Users can provide resources whenever they need without human interaction with the Provider

Broad Network Access

means

Services are accessible over the network through standard mechanisms and from various devices

Resource Pooling

means

Provider's resources are pooled to serve multiple consumers using a multi-tenant model with different physical and virtual resources

Rapid Elasticity

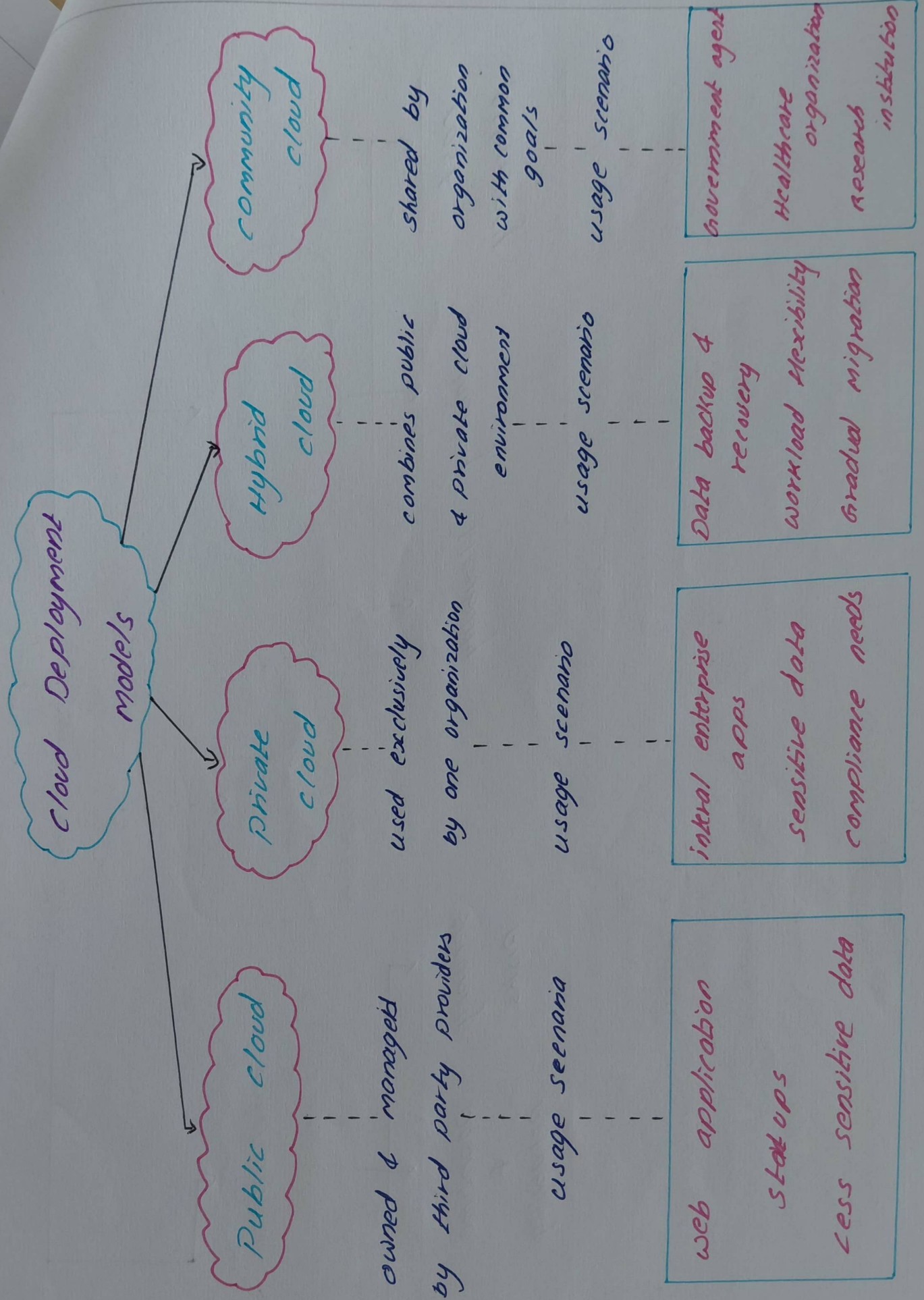
means

Resources can be up or down based on demand, appearing unlimited to the user

Measured Service

means

Cloud systems automatically optimize resource use by leveraging a metering capability



cloud computing

Benefits (Advantages)

cost
Efficiency

Reduces
Capital
Expenditure
and maintenance
costs. pay-
as-you-
use model.

scalability

Easily
scale
resources
up or
down based
on demand

Flexibility

supports
innovation
and
integration
with various
technologies

Balanced consideration

Challenges (Limitations)

Security
Risks

Data
breaches,
unauthorized
access and
compliance
concerns

Downtime
&
Reliability

outages
can
interrupt
services
and
affect
business
operation

Vendor
Lock-
in

Switching
providers
can
be difficult
and
costly